MAWLANA BHASHANI SCIENCE & TECHNOLOGY UNIVERSITY

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LAB REPORT

Department of : Computer Science and Engineering (CSE)
Lab Report No : 02
Lab Report On : Wave ploting, matrix operations, image processing, equation solving and highest and lowest pixel values for input images.
Course Tittle : Data and Communication Engineering
Course Code : CSE 2210

Submitted by,	Submitted to,
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Lab 2 (1): Create a sine wave and a cosine wave; Plot sinusoidal wave in a single graph using subplot.

Objective: Creating a sine wave and a cosine wave and plot sinusoidal wave in a single graph using subplot.

Required Software: MATLAB

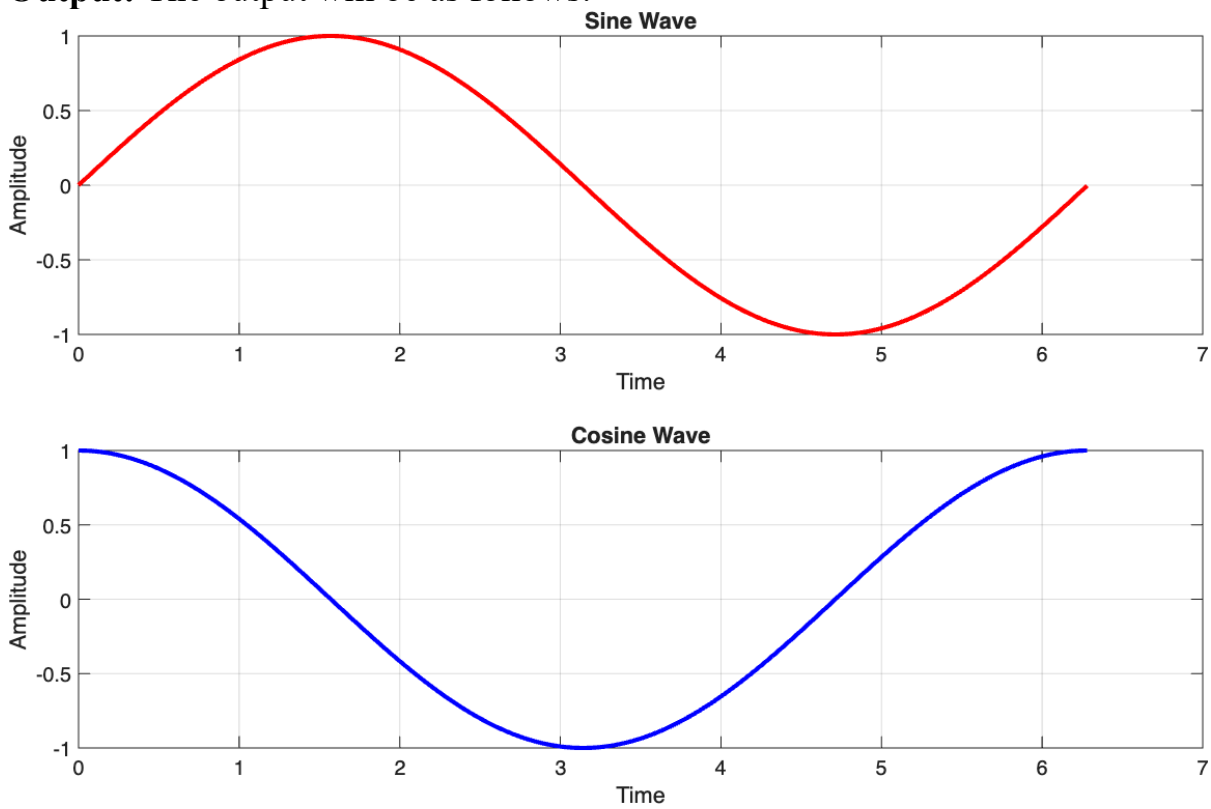
Code:

```
t = 0:0.01:2*pi;
s = sin(t);
c = cos(t);

subplot(2,1,1);
plot(t, s, 'r', 'LineWidth', 2);
title('Sine Wave');
xlabel('Time');
ylabel('Amplitude');
grid on;

subplot(2,1,2);
plot(t, c, 'b', 'LineWidth', 2);
title('Cosine Wave');
xlabel('Time');
ylabel('Amplitude');
grid on;
```

Output: The output will be as follows:



Lab 2 (2): Matrix operations with condition checking.

Objective: Creating matrix operations and checking the conditions.

Required Software: MATLAB

Code:

```
r = input('Enter row: ');
c = input('Enter column: ');

disp('Enter Matrix A:');
A = zeros(r,c);
for i = 1:r
    for j = 1:c
        A(i,j) = input('');
    end
end

disp('Enter Matrix B:');
B = zeros(r,c);
for i = 1:r
    for j = 1:c
        B(i,j) = input('');
    end
end

disp("A=")
disp(A)
disp("B=")
disp(B)

[r1, c1] = size(A);
[r2, c2] = size(B);

if isequal(size(A), size(B))
    add_res = A + B;
    sub_res = A - B;
    disp('Addition and Subtraction successful.');
else
    disp('!!Addition and Subtraction unsuccessful!!');
end

if c1 == r2
    mul_res = A * B;
    disp('Matrix Multiplication successful.');
else
    disp('!!Matrix Multiplication successful!!');
end
```

Output: The output will be as follows:

```
Enter row: 2
Enter column: 2
Enter Matrix A:
1
2
3
4
Enter Matrix B:
1
2
3
4
A=
     1     2
     3     4

B=
     1     2
     3     4

Addition and Subtraction successful.
Matrix Multiplication successful.
```

Lab 2 (3): Change Image format: png to jpg and others.

Objective: Changing an image into png to jpg and other image formats.

Required Software: MATLAB

Code:

```
[file, folderPath] = uigetfile({'*.png;*.jpg;*.jpeg;*.bmp', 'All Image
Files (*.png, *.jpg, *.bmp)'}, 'Select an image');

if isequal(file,0)
    disp('User cancelled the selection. Program terminated.');
else
    originalPath = fullfile(folderPath, file);
    img = imread(originalPath);

    disp('Available formats: jpg, bmp, tif, png, etc.');
    p = input('Which format do you want to save in? ', 's');
    outputName = ['converted_image.', p];

    imwrite(img, outputName);

    newImg = imread(outputName);
    figure;
    subplot(1, 2, 1);
    imshow(img);
    title(['Original: ', file]);

    subplot(1, 2, 2);
    imshow(newImg);
    title(['Converted Image (.', p, ')']);

    fprintf('\n--- Conversion Details ---\n');

    oldFileInfo = dir(originalPath);
    newFileInfo = dir(outputName);

    fprintf('Selected File: %s\n', fileName);
    fprintf('Original File Size: %.2f KB\n', oldFileInfo.bytes / 1024);
    fprintf('Converted File Size: %.2f KB\n', newFileInfo.bytes / 1024);

    newFileInfoDetail = imfinfo(outputName);
    fprintf('Confirmed Format in MATLAB: %s\n', newFileInfoDetail.Format);
    fprintf('Success! %s file has been saved.\n', outputName);
end
```

Output: The output will be as follows:

Original: MBSTULogo.png

Converted Image (.bmp)

Lab 2 (4): Solve simple equations.

Objective: Solving simple equations.

Required Software: MATLAB

Code:

```
syms x

eqn = x^2 - 5*x + 6 == 0;

sol = solve(eqn, x);

disp('The solutions are:');
disp(sol);
```

Output: The output will be as follows:

```
The solutions are:
2
3
```

Lab 2 (5): Find out highest and lowest pixel values for input images.

Objective: Finding the highest and lowest pixel values for input images.

Required Software: MATLAB

Code:

```
[file, path] = uigetfile({'.jpg;.png;*.bmp'}, 'Select an Image');
img = imread(fullfile(path, file));

if size(img, 3) == 3
    img = rgb2gray(img);
end

high_pixel = max(img(:));
low_pixel = min(img(:));

imshow(img);
title(['Max: ', num2str(high_pixel), ' | Min: ', num2str(low_pixel)]);

fprintf('File: %s\n', file);
fprintf('Highest Pixel Value: %d\n', high_pixel);
fprintf('Lowest Pixel Value: %d\n', low_pixel);
```

Output: The output will be as follows: